

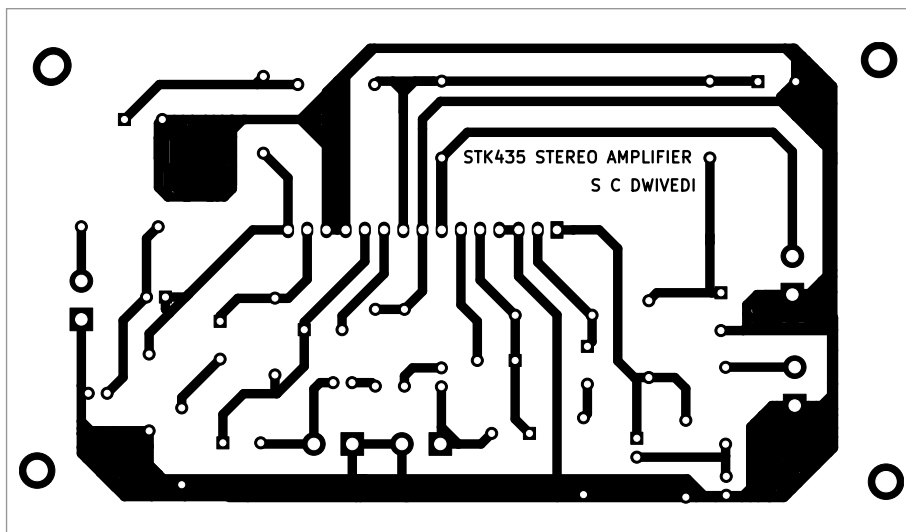


Fig. 3: Actual-size PCB layout

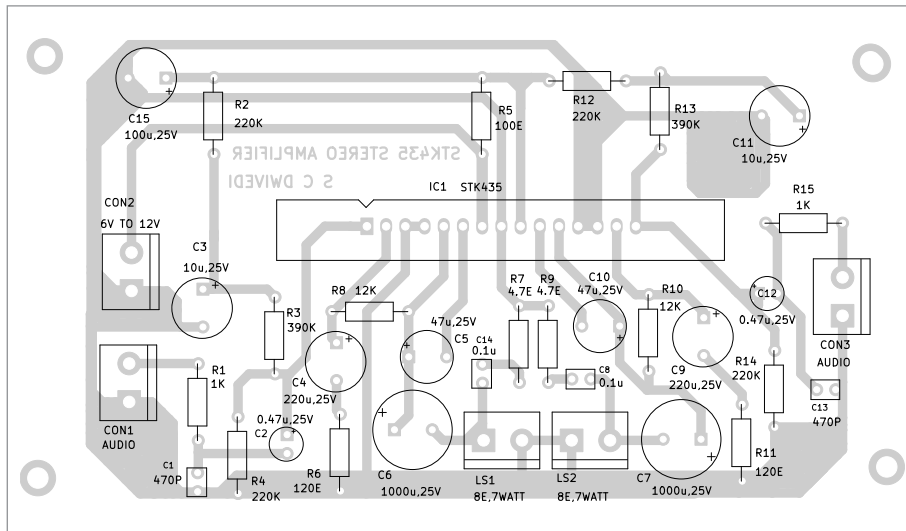


Fig. 4: Component layout

of IC1, with its negative terminal linked to a 1-kilo-ohm resistor (R1). Additionally, a 470pF ceramic capacitor (C1) connects the negative terminal of C2 to ground.

Pin 2 connects to the ground through an electrolytic capacitor of 220 μ F, 25V (C4) in series with a 120-ohm resistor (R6). Pins 3, 4, and 8 are also grounded. The output from pin 5 drives an 8-ohm, 7-watt speaker (LS1) via a 1000 μ F, 25V electrolytic capacitor (C6). The negative polarity of C6 connects to

ground through a series arrangement of a 0.1 μ F capacitor (C14) and a 4.7-ohm resistor (R7).

Pin 6 connects to pin 5 through a 47 μ F, 25V electrolytic capacitor (C5). Pin 7 is connected to the +12V supply, while pin 9 connects to +12V through a 100-ohm resistor (R5). A 12-kilo-ohm resistor (R8) bridges pins 2 and 5. A 100 μ F, 25V electrolytic capacitor (C15) connects pin 9 to ground.

The second amplifier section, IC1B, uses pins 10 through 15 and is

connected as shown in Fig. 2.

This circuit operates simply: connect audio inputs (from devices such as laptops, desktops, or MP3 players) to CON1 and CON3, then connect a 12V power supply. The amplified output will be heard through the speakers.

Construction and testing

An actual-size single-sided PCB layout for the STK435 amplifier is shown in Fig. 3, with the component layout in Fig. 4. After assembling the circuit on the PCB, enclose it in a suitable cabinet, ensuring proper heat dissipation for the IC with an appropriate heatsink.

Alternatively, a 7cm x 9cm Veroboard can be used. Position a 15-pin female connector in the middle of the board, two 2-pin connectors on the far right and left sides, and a 4-pin connector at the bottom. Solder all components as indicated in the circuit diagram, using a low-watt soldering iron. The IC should be inserted after soldering. Connect two 8-ohm, 7-watt speakers to the 2-pin connectors, and supply power using a 12V battery. A clicking sound from the speakers may indicate proper connection.

After assembling on either the PCB or a Veroboard, the device is ready for testing. Gently touch a metal screwdriver to the 2-pin connectors to hear a humming sound from the speakers, confirming functionality. Once tested, connect an MP3 player, laptop, or desktop to the amplifier's input and enjoy the audio output. **EFY**



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